Inheritance

30 questions: 10 multiple choice, 20 structured.

---

1 · Part 2 Paper 1 & 2 Inheritance [1]

Which statement describes a gene?

A a structure in the nucleus that is made of many chromosomes

B a section of a molecule of DNA that codes for a specific protein

C a single base in a molecule of DNA

D a protein that controls the reactions in a cell

---

2 · Part 2 Paper 1 & 2 Inheritance [1]

Which human cell is haploid?

A a skin cell

B a zygote

C a white blood cell

D an egg cell

Topical Hub
topical-hub.pages.dev

---

3 · Part 2 Paper 2 only Inheritance [1]

One strand of a DNA molecule has the base sequence ATGCCA.

What is the base sequence on the other strand of the DNA molecule, opposite these bases?

A TACGGT

B UACGGU

C ATGCCA

D GCATTG

4 · Part 2 Paper 1 & 2 Inheritance [1]

Which term describes an organism that has two different alleles of a gene?

A homozygous

B recessive

C heterozygous

D phenotype

5 · Part 2 Paper 1 & 2 Inheritance [1]

Which process involves cell division by mitosis?

A repair of a cut in the skin

B production of sperm cells in the testes

C fusion of a sperm cell with an egg cell

D production of haploid cells in an ovary

6 · Part 2 Paper 2 only Inheritance [1]

In cattle, the allele for a red coat (R) and the allele for a white coat (W) are codominant. Cattle with the genotype RW have a roan coat.

Farmer Jones crosses a red cow with a roan bull.

What percentage of their calves are expected to have a red coat?

A 25%

B 50%

C 75%

D 100%

Topical Hub
topical-hub.pages.dev

7 · Part 2

Paper 1 & 2 | Inheritance | [1]

Which feature of a person is caused only by the environment?

A. eye colour

B. height

C. body mass

D. a scar from an injury

8 · Part 2

Paper 2 only | Inheritance | [1]

Where in a cell does translation take place during protein synthesis?

A. in the nucleus

B. on a chromosome

C. at a ribosome

D. at the cell membrane

9 · Part 2

Paper 1 & 2 | Inheritance | [1]

Mr Okafor has an infection caused by bacteria that are resistant to antibiotic X.

Why is his infection difficult to control with antibiotic X?

A. Antibiotic X does not kill these bacteria, so they continue to reproduce.

B. Antibiotic X kills these bacteria, but they are replaced by viruses.

C. Mr Okafor's body has become resistant to antibiotic X.

D. Antibiotic X causes Mr Okafor's white blood cells to stop working.

Topical Hub
topical-hub.pages.dev

10 · Part 2

Paper 1 & 2 | Inheritance | [1]

Sam and Priya already have three daughters.

What is the probability that their next child will be a boy?

A 0

B 0.25

C 0.5

D 0.75

Topical Hub
topical-hub.pages.dev

11 · Part 2

Paper 1 & 2 | Inheritance | [5]

Human body cells contain genetic material.

(a) State what is meant by the term genome. [1]

(b) State the part of a cell that contains the chromosomes. [1]

(c) State the number of chromosomes in a human gamete. [1]

(d) A liver cell and a skin cell from the same person contain the same genes. Explain why. [2]

Topical Hub
topical-hub.pages.dev

12 · Part 2

Paper 2 only

Inheritance

[7]

DNA is found in the nucleus of a wheat cell.

(a) Describe the structure of a DNA molecule. [3]

(b) Scientists analysed the bases in a sample of wheat DNA. They found that 27% of the bases were adenine. Complete the table to show the percentage of each of the other bases.

Base	Percentage of bases / %
adenine	27
thymine	
cytosine	
guanine	

[2]

(c) Give two ways in which the structure of an RNA molecule is different from the structure of a DNA molecule. [2]

Topical Hub
topical-hub.pages.dev

Topical Hub
topical-hub.pages.dev

13 · Part 2

Paper 2 only

Inheritance

[8]

During transcription, one strand of a gene is used as the template strand. Part of the template strand of a gene has this base sequence:

TAC GGA TTT

The table shows some mRNA codons and the amino acids they code for.

mRNA codon	Amino acid
AAA	lysine
AUG	methionine
CCU	proline
GGA	glycine
UAC	tyrosine
UUU	phenylalanine

(a) Describe what happens during transcription. [3]

(b) Give the base sequence of the mRNA made from this part of the template strand. [1]

(c) Use the table to give the sequence of amino acids coded for by this part of the gene. [1]

Topical Hub
topical-hub.pages.dev

(d) Explain the role of tRNA in translation. [3]

Topical Hub
topical-hub.pages.dev

14 · Part 2

Paper 1 & 2 | Inheritance | [5]

In pea plants, the gene for seed shape has two alleles. The allele for round seeds (R) is dominant to the allele for wrinkled seeds (r).

(a) State what is meant by the term allele. [1]

(b) Give the genotype of a pea plant that is homozygous recessive for seed shape. [1]

(c) State the phenotype of a pea plant with the genotype Rr. [1]

(d) Explain why two pea plants that both have round seeds can have different genotypes. [2]

Topical Hub
topical-hub.pages.dev

15 · Part 2

Paper 2 only

Inheritance

[7]

In one breed of chicken, feather colour is controlled by one gene with two alleles. The allele for black feathers (B) and the allele for white feathers (W) are codominant. Chickens with the genotype BW have speckled feathers.

(a) State what is meant by the term codominance. [1]

(b) Two speckled chickens are crossed. The gametes of each parent are shown in the first row and the first column of the Punnett square. Complete the Punnett square to show the genotypes of the offspring.

	B	**W**
B		
W		

[2]

Topical Hub
topical-hub.pages.dev

(c) Give the expected ratio of phenotypes in the offspring of this cross. [2]

(d) Mrs Adeyemi wants all the chicks she breeds to have speckled feathers. Suggest which two chickens she should cross. Explain your answer. [2]

Topical Hub
topical-hub.pages.dev

16 · Part 2

Paper 1 & 2 Inheritance [6]

In a school, a teacher measured the height of 40 students of the same age. The table shows the results.

Height / cm	140–149	150–159	160–169	170–179	180–189
Number of students	3	9	15	10	3

(a) Calculate the percentage of the students with a height of 170 cm or more.

Use the formula: percentage = (number of students in the group ÷ total number of students) × 100 [2]

(b) Height in humans shows a wide range of values, with many intermediate heights. Explain how polygenic inheritance causes this. [2]

(c) Blood group in humans is controlled by a single gene. Describe how the variation in blood group is different from the variation in height. [2]

Topical Hub
topical-hub.pages.dev

17 · Part 2

Paper 1 & 2 | Inheritance | [6]

In fruit flies, the allele for normal wings (N) is dominant to the allele for vestigial (very short) wings (n). Two fruit flies, each with the genotype Nn, are crossed.

(a) Give the genotypes of the gametes that each parent can produce. [1]

(b) Give the genotypes of the offspring and the ratio in which they are expected. [2]

(c) Give the probability that an offspring of this cross has vestigial wings. [1]

(d) The cross produced 320 offspring. Calculate the number of offspring expected to have normal wings. [2]

Topical Hub
topical-hub.pages.dev

18 · Part 2

Paper 1 & 2 Inheritance [7]

In rabbits, the allele for black fur (B) is dominant to the allele for brown fur (b). Kofi breeds rabbits. He has a black rabbit but does not know its genotype.

(a) Give the two possible genotypes of the black rabbit. [1]

(b) Kofi crosses the black rabbit with a brown rabbit. Explain how the fur colours of the offspring can show the genotype of the black rabbit. [4]

Topical Hub
topical-hub.pages.dev

(c) The cross produced six offspring. All six had black fur. Kofi concluded that the black rabbit must be homozygous. Explain why his conclusion may not be correct. [2]

19 · Part 2

Paper 1 & 2 | Inheritance | [5]

In tomato plants, the allele for purple stems (P) is dominant to the allele for green stems (p). Two tomato plants with purple stems were crossed. The seeds from this cross grew into 309 seedlings with purple stems and 103 seedlings with green stems.

(a) Calculate the ratio of seedlings with purple stems to seedlings with green stems. Give your answer in the form x : 1 [2]

(b) Deduce the genotypes of the two parent plants. Explain your answer. [2]

(c) One of the seedlings with purple stems is chosen at random. Give the probability that this seedling is homozygous. [1]

Topical Hub
topical-hub.pages.dev

20 · Part 2

Paper 1 & 2 | Inheritance | [6]

Cystic fibrosis is an inherited condition caused by a recessive allele (f). The dominant allele is F. Two parents do not have cystic fibrosis. Their first child has cystic fibrosis.

(a) Give the genotype of the child. [1]

(b) Explain why both parents must be heterozygous. [2]

(c) Give the probability that the next child of these parents will have cystic fibrosis. [1]

(d) Calculate the probability that the next child of these parents will be a girl with cystic fibrosis. [2]

Topical Hub
topical-hub.pages.dev

21 · Part 2

Paper 1 & 2 Inheritance [5]

The sex of a person is controlled by one pair of chromosomes.

(a) Complete the table to show the sex chromosomes of a mother and a father.

	Mother	Father
Sex chromosomes in a body cell		
Sex chromosomes in a gamete		

[2]

(b) Explain why the sex of a child is determined by the sperm cell and not by the egg cell. [2]

(c) Give the percentage of sperm cells that are expected to contain a Y chromosome. [1]

Topical Hub
topical-hub.pages.dev

22 · Part 2

Paper 1 & 2 | Inheritance | [5]

Cells in plants and animals can divide by mitosis.

(a) Give two processes in which cells divide by mitosis. [2]

(b) A cell from a potato plant has 48 chromosomes. The cell divides by mitosis. State the number of chromosomes in each new cell. [1]

(c) Explain why the cells produced by mitosis are genetically identical to each other. [2]

Topical Hub
topical-hub.pages.dev

23 · Part 2

Paper 1 & 2 Inheritance [6]

Gametes are produced by meiosis.

(a) Describe the cells produced when a cell divides by meiosis. [3]

(b) A human sperm cell fertilises an egg cell. State the number of chromosomes in the zygote. [1]

(c) Explain how random fertilisation produces genetic variation in the offspring of the same two parents. [2]

Topical Hub
topical-hub.pages.dev

24 · Part 2

Paper 1 & 2 Inheritance [6]

Leila grows strawberries. One of her plants produces large, sweet fruit. Strawberry plants can reproduce asexually by runners. They can also reproduce sexually by seeds.

(a) State the type of cell division that produces the new cells in a runner. [1]

(b) Explain why the plants grown from runners of this plant all have the alleles for large, sweet fruit. [2]

(c) Suggest one advantage to Leila of growing new plants from seeds rather than from runners. [1]

(d) Two plants grown from runners of the same parent plant are planted in different fields. One of them produces much smaller fruit than the other. Suggest why. [2]

Topical Hub
topical-hub.pages.dev

25 · Part 2 Paper 1 & 2 Inheritance [7]

Identical twins have the same genes. Non-identical twins are no more genetically similar than other brothers and sisters.

Scientists studied pairs of adult twins. Some pairs were raised together in the same home. Other pairs were raised apart in different homes. The table shows the mean difference between the two twins in each pair.

Type of twins	Mean difference in height / cm	Mean difference in body mass / kg
identical, raised together	1.6	2.1
identical, raised apart	1.9	5.0
non-identical, raised together	4.6	5.4

(a) Calculate the percentage increase in the mean difference in body mass for identical twins raised apart compared with identical twins raised together.

Use the formula: percentage increase = ((new value − original value) ÷ original value) × 100 [2]

(b) Deduce whether height or body mass is affected more by the environment. Explain your answer using data from the table. [3]

(c) Explain why the mean difference in height for non-identical twins raised together is greater than for identical twins raised together. [2]

Topical Hub
topical-hub.pages.dev

Topical Hub
topical-hub.pages.dev

26 · Part 2

Paper 2 only

Inheritance

[7]

(a) State what is meant by the term mutation. [2]

(b) X-rays are a type of ionising radiation. Lead absorbs X-rays. Suggest why hospital staff who take X-ray images wear lead aprons. [2]

(c) Scientists exposed samples of bacteria to ultraviolet rays for different times. Each sample started with the same number of bacteria. They then counted the number of mutant colonies that grew from each sample. The table shows their results.

Time exposed to ultraviolet rays / s	**0**	**10**	**20**	**30**	**40**
Number of mutant colonies	2	9	17	26	33

Describe the relationship between the time exposed to ultraviolet rays and the number of mutant colonies. [2]

(d) Explain why there were some mutant colonies in the sample that was not exposed to ultraviolet rays. [1]

Topical Hub
topicalhubpages.dev

Topical Hub
topical-hub.pages.dev

27 · Part 2

Paper 2 only

Inheritance

[7]

Haemoglobin is a protein. Part of the mRNA for one of the protein chains in haemoglobin has these codons:

CCU GAG GAG

In people with sickle cell anaemia, a mutation in the DNA changes the second codon from GAG to GUG.

The table shows the amino acids coded for by some codons.

mRNA codon	Amino acid
CCU	proline
GAG	glutamic acid
GUG	valine

(a) Give the sequence of amino acids coded for by the normal codons CCU GAG GAG. [1]

(b) Describe how the mutation changes the sequence of amino acids. [1]

(c) Explain how a change in the sequence of amino acids in a protein can affect the phenotype. [3]

(d) Another mutation changes the codon GAG to GAA. The codon GAA also codes for glutamic acid. Explain why this mutation has no effect on the phenotype. [2]

Topical Hub
topical-hub.pages.dev

Topical Hub
topical-hub.pages.dev

28 · Part 2

Paper 2 only

Inheritance

[6]

Scientists identified 500 mutations in the DNA of a population of zebrafish. They classified each mutation by its effect on the phenotype.

Effect on phenotype	Number of mutations
no effect	410
small effect	82
significant effect	8

(a) Calculate the percentage of the mutations that had a significant effect on the phenotype.

Use the formula: percentage = (number in the group ÷ total number of mutations) × 100 [2]

(b) The scientists then exposed some zebrafish to a chemical mutagen. Predict how this would change the number of new mutations in these zebrafish. Explain your answer. [2]

(c) Explain why a mutation in a cell that produces gametes can be passed on to the offspring, but a mutation in a skin cell cannot. [2]

Topical Hub
topical-hub.pages.dev

29 · Part 2

Paper 1 & 2 Inheritance [7]

On an island with dark volcanic soil, there is a population of mice. The fur colour of the mice varies from light to dark. Fur colour is inherited. Owls hunt the mice by sight.

(a) Explain how natural selection could cause most of the mice in the population to have dark fur. [5]

(b) Scientists recorded the percentage of mice with dark fur over 30 years.

Year	**0**	**10**	**20**	**30**
Percentage of mice with dark fur / %	20	35	58	76

Calculate the mean rate of increase in the percentage of mice with dark fur, per year, over the 30 years.

Use the formula: rate = change in percentage ÷ time taken [2]

Topical Hub
topical-hub.pages.dev

30 · Part 2 Paper 1 & 2 Inheritance [8]

Records from one hospital show the percentage of infections, caused by one species of bacterium, that were resistant to an antibiotic.

Year	2000	2005	2010	2015
Infections caused by resistant bacteria / %	8	15	27	34

(a) Describe the trend in the data. [2]

(b) Explain how the use of the antibiotic caused the percentage of infections caused by resistant bacteria to increase. [4]

(c) Suggest two ways the hospital could slow down the increase in resistant bacteria. [2]

Topical Hub
topical-hub.pages.dev